

Flagged LLT polynomials and nonsymmetric Macdonald polynomials

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UC Davis Algebra & Discrete Mathematics Seminar

joint work with Jonah Blasiak, Mark Haiman, Jennifer Morse, and Anna Pun

20 April 2026

Symmetric polynomials and functions

- Symmetric polynomials $\mathbb{Q}[x_1, \dots, x_n]^{S_n}$
- Generators

$$e_r(x_1, \dots, x_n) = \sum_{1 \leq i_1 < i_2 < \dots < i_r \leq n} x_{i_1} \cdots x_{i_r}$$

$$e_1(x_1, x_2, x_3) = x_1 + x_2 + x_3$$

$$e_2(x_1, x_2, x_3) = x_1x_2 + x_1x_3 + x_2x_3$$

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- Integer partitions of d .

Partitions

Definition

$n \in \mathbb{Z}_{>0}$, a *partition of n* is $\lambda = (\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\ell > 0)$ such that $\lambda_1 + \lambda_2 + \cdots + \lambda_\ell = n$.

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5 $\rightarrow$

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$4 + 1 \rightarrow$ 

$$3 + 2 \rightarrow \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}$$

$3 + 1 + 1 \rightarrow$ 

$$2 + 2 + 1 \rightarrow \begin{array}{|c|c|} \hline \square & \square \\ \hline \square & \square \\ \hline \square & \square \\ \hline \end{array}$$

$$2 + 1 + 1 + 1 \rightarrow \begin{array}{|c|} \hline \square \\ \hline \square \\ \hline \square \\ \hline \square \\ \hline \end{array}$$

$$1 + 1 + 1 + 1 + 1 \rightarrow$$

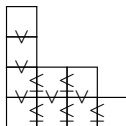
Young Tableaux

Definition

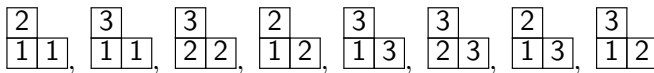
Filling of partition diagram of λ with numbers such that

- 1 strictly increasing up columns
- 2 weakly increasing along rows

Collection is called $\text{SSYT}(\lambda)$.



For $\lambda = (2, 1)$,



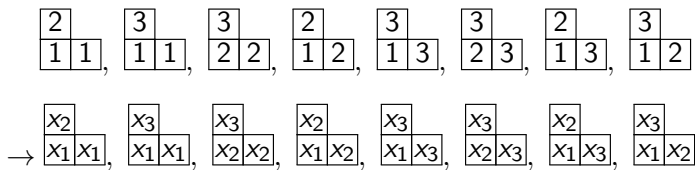
Schur polynomials

Associate a polynomial to $\text{SSYT}(\lambda)$.

2		3		3		2		3		3		2		3	
1	1	1	1	2	2	1	2	1	3	2	3	1	3	1	2

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$$s_{(2,1)}(x_1, x_2, x_3) = x_1^2 x_2 + x_1^2 x_3 + x_2^2 x_3 + x_1 x_2^2 + x_1 x_3^2 + x_2 x_3^2 + 2x_1 x_2 x_3$$

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Definition

For λ a partition

$$s_\lambda = \sum_{T \in \text{SSYT}(\lambda)} x^T \text{ for } x^T = \prod_{i \in T} x_i$$

- s_λ is a symmetric function.

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- s_λ is a symmetric function.
- $\{s_\lambda\}_\lambda$ forms a basis for Λ .

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Hidden Guide: Schur Positivity

“Naturally occurring” symmetric functions which are non-negative (coefficients in $\mathbb{N}$) linear combinations in Schur polynomial basis are interesting since they could have representation-theoretic models.

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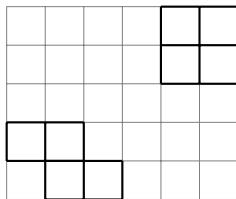
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- q -deformation?

Key Object: LLT Polynomials

Let $\nu = (\nu_{(1)}, \dots, \nu_{(k)})$ be a tuple of skew shapes. (Skew shape = $\lambda \setminus \mu$)

$$\nu = \left(\begin{array}{|c|c|c|} \hline \square & \square & \\ \hline \square & \square & \square \\ \hline \end{array}, \begin{array}{|c|c|c|} \hline \square & \square & \square \\ \hline \square & \square & \square \\ \hline \end{array} \right)$$



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-4	-3	-2	-1	0	1
-3	-2	-1	0	1	2
-2	-1	0	1	2	3
-1	0	1	2	3	4
0	1	2	3	4	5

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Attacking pairs: $(b_2, b_3), (b_3, b_4), (b_4, b_5), (b_4, b_6), (b_5, b_7), (b_6, b_7), (b_7, b_8)$

Key Object: LLT Polynomials

Let $\nu = (\nu_{(1)}, \dots, \nu_{(k)})$ be a tuple of skew shapes. (Skew shape $= \lambda \setminus \mu$)

- The *content* of a box in row y , column x is $x - y$.
- *Reading order*: label boxes $b_1, \dots, b_n$ by scanning each diagonal from southwest to northeast, in order of increasing content.
- A pair $(a, b) \in \nu$ is *attacking* if a precedes b in reading order and
 - $\text{content}(b) = \text{content}(a)$, or
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LLT Polynomials

- A *semistandard tableau* on ν is a map $T: \nu \rightarrow \mathbb{Z}_+$ which restricts to a semistandard tableau on each $\nu_{(i)}$.

The *LLT polynomial* indexed by a tuple of skew shapes ν is

$$\mathcal{G}_\nu(x; q) = \sum_{T \in \text{SSYT}(\nu)} x^T,$$

$$x^T = \prod_{a \in \nu} x_{T(a)}.$$

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$$\text{inv}(T) = 4, \quad x^T = x_1^2 x_2 x_3 x_4 x_5^2 x_6$$

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- Thus, $\mathcal{G}_\nu$ are useful to show combinatorial quantities are Schur-positive!

Some notable occurrences of LLT Polynomials

- Haglund-Haiman-Loehr formula for Macdonald polynomials:

$$\tilde{H}_{\mu}(x; q, t) = t^{n(\mu)} \sum_R \left(\prod_{\substack{\square \\ u}} q^{a+1} t^l \right) \mathcal{G}_R(x; t^{-1}).$$

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- For $G =$ incomparability graph for natural unit interval order (encoded by Dyck path P), the t -chromatic symmetric polynomial is a “signed” LLT polynomial (later), i.e.,

$$\chi_G(x; t) = (1 - t)^{-|V|} \mathcal{G}_{\nu(P)}[(1 - t)x; t].$$

Flagged Tableaux

- Fix partition shape $\lambda = (\lambda_1, \dots, \lambda_l)$ and *flag*
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- A *flagged semistandard Young tableau* of shape λ with flag $\mathbf{b}$ is a $T \in \text{SSYT}(\lambda)$ such that the entries of row i are bounded above by b_i .
- Denote the set of such tableaux via $\text{FT}(\lambda, \mathbf{b})$.
- E.g., $\lambda = (2, 1)$, $\mathbf{b} = (1, 3)$, $\text{FT}(\lambda, \mathbf{b}) =$

$$\begin{array}{|c|c|} \hline 2 & \\ \hline 1 & 1 \\ \hline \end{array} \leq 1 \quad \begin{array}{|c|c|} \hline 3 & \\ \hline 1 & 1 \\ \hline \end{array} \leq 1$$

Flagged Schur functions (Lascoux-Schützenberger, Wachs)

- For partition λ and flag $\mathbf{b}$, the *flagged Schur function* is given by

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- $s_{\lambda, \mathbf{b}}(x)$ are examples of *Demazure characters*.

Demazure characters and atoms

The *Demazure operator* π_i acts on $f \in \mathbb{Q}(q, t)[x_1^{\pm 1}, \dots, x_N^{\pm 1}]$ by

$$\pi_i(f) = \frac{x_i f - x_{i+1} s_i(f)}{x_i - x_{i+1}}.$$

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The *Demazure characters* or *key polynomials* are constructed from

- $\mathcal{D}_\lambda = x^\lambda := x_1^{\lambda_1} \cdots x_N^{\lambda_N}$ for partition λ .
- $\mathcal{D}_{s_i(\alpha)} = \pi_i \mathcal{D}_\alpha$ for $\alpha_i > \alpha_{i+1}$, for any $\alpha \in \mathbb{N}^N$.

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Demazure atoms are defined the same as keys but with $\hat{\pi}_i := \pi_i - 1$ in place of π_i :

- $\mathcal{A}_\lambda = x^\lambda$ for partition λ .
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Examples

$$\mathcal{D}_{520} = x_1^5 x_2^2$$

$$\mathcal{D}_{250} = \pi_1 \mathcal{D}_{520} = \pi_1(x_1^5 x_2^2) = x_1^5 x_2^2 + x_1^4 x_2^3 + x_1^3 x_2^4 + x_1^2 x_2^5$$

$$\mathcal{D}_{205} = \pi_2 \mathcal{D}_{250} = \pi_2(x_1^5 x_2^2 + x_1^4 x_2^3 + x_1^3 x_2^4 + x_1^2 x_2^5)$$

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$$\mathcal{D}_{520} = \mathcal{A}_{520}$$

$$\mathcal{D}_{250} = \mathcal{A}_{520} + \mathcal{A}_{250}$$

$$\mathcal{D}_{205} = \mathcal{A}_{520} + \mathcal{A}_{250} + \mathcal{A}_{502} + \mathcal{A}_{205}$$

$$\vdots$$

Recovering Symmetric Functions

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- For *Weyl symmetrization operator* π_{w_0} , we have

$$\pi_{w_0}(s_{\lambda, \mathbf{b}}(x)) = s_{\lambda}(x), \quad \pi_{w_0}(\mathcal{D}_{\alpha}) = \mathcal{D}_{\text{sort}(\alpha)} = s_{\alpha+},$$

$$\pi_{w_0}(\mathcal{A}_{\alpha}) = \begin{cases} s_{\alpha} & \alpha \text{ a partition,} \\ 0 & \text{else.} \end{cases}$$

Flagged LLT Polynomials (Blasiak-Haiman-Morse-Pun-S.)

- Let $e_1, \dots, e_l$ be the row ends of ν , ordered in reverse reading order.
- Fix flag $\mathbf{b} = (b_1 \leq \dots \leq b_l)$.
- $\text{FT}(\nu, \mathbf{b})$ = set of semistandard tableaux T on ν satisfying $T(e_i) \leq b_i$.
- The *flagged LLT polynomial* indexed by ν and $\mathbf{b}$ is

$$\mathcal{G}_{\nu, \mathbf{b}}(x; t) = \sum_{T \in \text{FT}(\nu, \mathbf{b})} t^{\text{inv}(T)} x^T.$$

						e_3
						e_1
	e_4					
		e_2				

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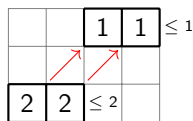
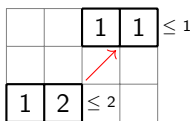
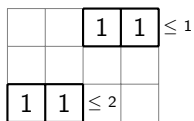
$$\mathcal{G}_{\nu, \mathbf{b}}(x; t) = \sum_{T \in \text{FT}(\nu, \mathbf{b})} t^{\text{inv}(T)} x^T.$$

				2	3	≤ 3
				1	1	≤ 1
2	4					≤ 4
	1	2				≤ 2

Flagged LLT Polynomials

$$\nu = (\square\square, \square\square), \mathbf{b} = (1, 2)$$

T



$$\mathcal{G}_{r,\nu}(x; t) = x_1^4 + t x_1^3 x_2 + t^2 x_1^2 x_2^2$$

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- $\mathcal{G}_{\nu, \mathbf{b}}(x; t)$ are conjecturally positive in terms of *Demazure atoms*.

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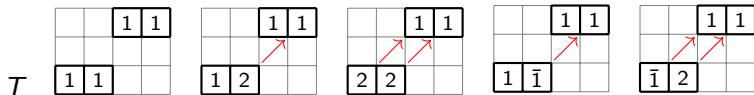
The *signed flagged LLT polynomial* indexed by ν and flag $\mathbf{b}$ is

$$\mathcal{G}_{\nu, \mathbf{b}}^\pm(x; t) = \sum_{T \in \text{FT}^\pm(\nu, \mathbf{b})} t^{\text{inv}(T)} (-t)^{-\#\text{bar}(T)} x^{|T|},$$

where $|T|$ is the result of removing all bars from T .

Signed flagged LLT polynomials

$$\nu = (\square\square, \square\square), \mathbf{b} = (1, 2)$$



$$\begin{aligned} \mathcal{G}_{\nu, \mathbf{b}}^{\pm}(x; t) &= x_1^4 + tx_1^3x_2 + t^2x_1^2x_2^2 - x_1^4 - tx_1^3x_2 \\ &= t^2x_1^2x_2^2 \end{aligned}$$

Flagged plethysm

Define *flagged plethysm* $\Pi_{t,x}: \mathbb{k}[x_1, \dots, x_l] \rightarrow \mathbb{k}[x_1, \dots, x_l]$ to be the (over-determined) linear map

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Theorem (Blasiak-Haiman-Morse-Pun-S., 2025+)

$\Pi_{t,x}$ is well-defined.

$\Pi_{t,x}$ is a “nonsymmetric analogue” of the plethystic map $f[X] \mapsto f[X/(1-t)]$ for symmetric function f .

Applications

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- ② A “nonsymmetric shuffle theorem” expressed in terms of flagged LLT polynomials associated to flagged Dyck paths.
- ③ Tewari-Wilson-Zhang define chromatic nonsymmetric polynomials associated to $d \times d$ Dyck paths starting with r north steps, $\chi_{\mathbf{b}, \pi}$. This is (essentially) a signed flagged LLT polynomial.

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- Plethystically modified Macdonald polynomials

$$H_\mu(x; q, t) = J_\mu[x/(1-t); q, t].$$

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$$H_{31} = ts_4 + (1 + qt + q^2t)s_{31} + (q + tq^2)s_{22} + (q + q^2 + q^3t)s_{211} + q^3s_{1111}$$

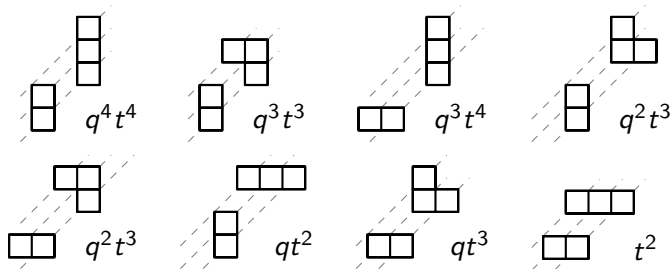
Haglund-Haiman-Loehr formula example

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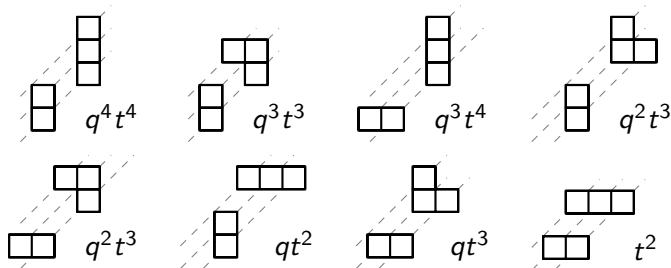
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Remark: Can permute the order of ribbons (but changes arm and leg statistics and LLTs.)

Nonsymmetric Macdonald polynomials

The Cherednik operators $Y_1, \dots, Y_N$ act on $\mathbb{Q}(q, t)[x_1^{\pm 1}, \dots, x_N^{\pm 1}]$.

$$T_i f = s_i f + (1 - t) x_i \frac{f - s_i f}{x_i - x_{i+1}}, \quad (\text{Demazure-Lusztig operators})$$

$$\Phi f = f(x_2, \dots, x_N, qx_1),$$

$$Y_i = t^{-i+1} T_{i-1} \cdots T_1 x_1 \Phi T_{N-1}^{-1} \cdots T_i^{-1}.$$

Definition

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- The $\mathcal{E}_\alpha$'s “Hecke symmetrize” to the integral form J_μ 's.

Nonsymmetric HHL formula example

$$\mathcal{E}_\alpha(x; q, t) = t^{n(\mu_+)} \sum_R \left(\prod_{\substack{u \\ \square}} q^{a+1} t^l \right) \mathcal{G}_{R, (1, 2, \dots, m)}^\pm(x; t^{-1}).$$

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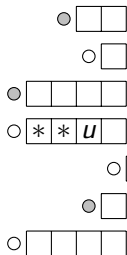
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For $\alpha = (2, 0, 1)$, we get 8 signed flagged LLT terms:

$\begin{array}{ c c } \hline \square & \square \\ \hline \end{array} 1$	$\begin{array}{ c c } \hline \square & \square \\ \hline \end{array} 1$	$\begin{array}{ c } \hline \square \\ \hline \square \\ \hline \end{array} 1$	$\begin{array}{ c } \hline \square \\ \hline \square \\ \hline \end{array} 1$
$t \quad 2$	$q t \quad 2$	$q t^2 \quad 2$	$q^2 t^2 \quad 2$
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$q^2 t^3 \quad 2$	$q^3 t^3 \quad 2$	$q^3 t^4 \quad 2$	$q^4 t^4 \quad 2$
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- Convention: bound boxes are always on the same content “diagonal.”
- Variation: bound boxes below shape bound shape strictly from below.

Arms and legs



$$a(u) = 2, l(u) = 3$$

The Missing Corner

$$\begin{array}{ccc} \mathcal{E}_\alpha(x_1, \dots, x_N; q, t) & \longrightarrow & ? \\ \downarrow \text{Hecke sym.} & & \downarrow \\ J_\mu(x; q, t) & \xrightarrow{\text{Plethysm}} & H_\mu(x; q, t) \end{array}$$

Question

Is there a “nice” family of polynomials that Weyl symmetrizes to the modified Macdonald polynomials?

An important feature for “nice”? Positivity!

Toward positivity

- Knop (2007) formulated a positivity conjecture for a stable version of $\mathcal{E}_\alpha$ involving Kazhdan-Lusztig theory.
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- We can understand stabilizations in terms of signed flagged LLT combinatorics.
- Upshot: look at stabilizations for “positivity.”

Filling in the missing corner

- $\mathcal{P}(r) = \mathbb{Q}(q, t)[x_1, \dots, x_r] \otimes \Lambda_{\mathbb{Q}(q, t)}(x_{r+1}, \dots)$.
- Use a “stable” version of $\mathcal{E}_\alpha$, which we denote $\mathcal{J}_{\eta|\lambda}$.

$$\begin{array}{ccc}
 \mathcal{J}_{\eta|\lambda}(x; q, t) & \xrightarrow{\Pi_r} & \text{nsH}_{\eta|\lambda}(x; q, t) \\
 \downarrow \text{Hecke sym.} & & \downarrow \text{Weyl sym.} \\
 J_{(\eta; \lambda)_+}(x; q, t) & \xrightarrow{\text{Plethysm}} & H_{(\eta; \lambda)_+}(x; q, t)
 \end{array}$$

- $\mathcal{J}_{\eta|\lambda}(x; q, t)$ and $\text{nsH}_{\eta|\lambda}(x; q, t)$ for $(\eta|\lambda) \in \mathbb{N}^r \times \text{Par}$ form bases of $\mathcal{P}(r)$.
- $(\eta; \lambda)_+ =$ partition rearrangement of concatenating $(\eta; \lambda)$.
- Π_r is “ r -nonsymmetric plethysm”

$$\Pi_r(f(x_1, \dots, x_r) \otimes g(x)) = (\Pi_{t,x} f(x_1, \dots, x_r)) \otimes g[x/(1-t)] .$$

Modern Macdonald polynomials

Definition

The *modern Macdonald polynomial* indexed by $\eta \in \mathbb{N}^r$ and partition λ is

$$\text{nsH}_{\eta|\lambda}(x; q, t) = t^{n((\eta;\lambda)_+)} \sum_R \left(\prod_{\substack{u \\ \square}} q^{a+1} t^l \right) \mathcal{G}_{R, (1,2,\dots,r)}(x; t^{-1})$$

For $r = 2$, $\eta = (2, 1)$, $\lambda = \emptyset$, the LLT terms are

$$\begin{array}{cc}
 \begin{array}{c} \square \square \quad 1 \\ t \\ \square \quad 2 \end{array} &
 \begin{array}{c} \square \\ q t^2 \\ \square \quad 2 \end{array}
 \end{array}$$

$$\text{nsH}_{21|\emptyset}(x_1, x_2, x_3; q, t) = t x_1^3 + x_1^2 x_2 + q t x_1^2 x_2 + q t x_1 x_2^2 + q t x_1^2 x_3 + q x_1 x_2 x_3$$

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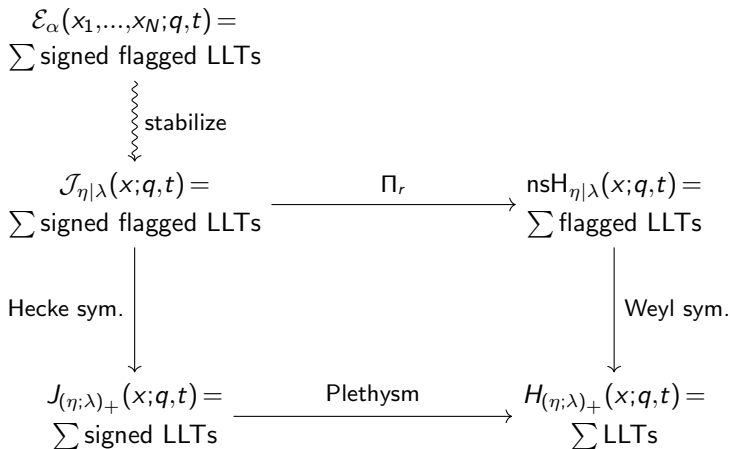
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- $\text{nsH}_{\eta|\lambda}(x; q, t)$ form a basis of $\mathcal{P}(r)$.
- We conjecture $\text{nsH}_{\eta|\lambda}(x; q, t)$ is atom positive (would follow from flagged LLTs being atom positive).

Modern Macdonald summary



Extra slides

What follows are extra slides that were not presented.

Stabilization

Definition

A sequence $g_1, g_2, \dots$ with $g_N \in \mathbb{Q}(q, t)[x_1, \dots, x_N]$ converges t -adically to $f(x) \in \mathcal{P}(r)$ if, for all $e \geq 0$,

$$g_N(x_1, \dots, x_N) - f(x_1, \dots, x_N, 0, 0, \dots)$$

has coefficients whose order of vanishing in t is at least e , for sufficiently large N .

- $1, 1 + t, 1 + t + t^2, \dots \rightarrow \frac{1}{1-t}$.
- $x_1, tx_1 + x_2, t^2x_1 + x_2 + x_3, t^3x_1 + x_2 + x_3 + x_4, \dots \rightarrow x_2 + x_3 + x_4 + \dots$.

Stable nonsymmetric Macdonald polynomials

Definition

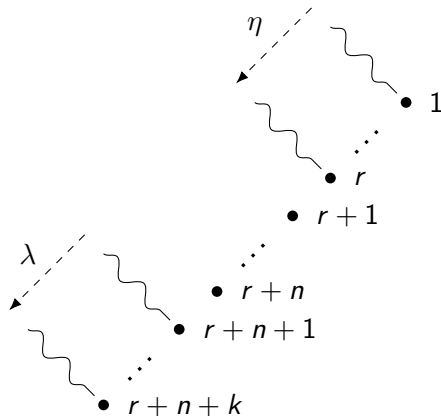
For $(\eta|\lambda) \in \mathbb{N}^r \times \text{Par}$, the *integral form stable r -nonsymmetric Macdonald polynomial* $\mathcal{J}_{\eta|\lambda}(x; q, t) \in \mathcal{P}(r)$ is given by

$$\mathcal{J}_{\eta|\lambda}(x; q, t) = \lim_{n \rightarrow \infty} \mathcal{E}_{(\eta; 0^n; \lambda)}(x_1, \dots, x_{r+n}, 0^{\ell(\lambda)}; q, t).$$

These $\mathcal{J}_{\eta|\lambda}$'s are integral forms of stable versions of E_α introduced by Bechtloff Weising.

Stable nonsymmetric Macdonald polynomials

Fix some $r \in \mathbb{Z}_{>0}$. Add n empty ribbons after the first r . Set variables $x_{r+n} = x_{r+n+1} = \cdots = 0$.



On (signed) flagged ribbon LLTs, this causes bounds $\geq r+n+1$ to be no restriction and ribbons with bounds below ribbon $\rightarrow 0$.

The plethystic relationship

Theorem (Blasiak-Haiman-Morse-Pun-S., 2025+)

$$\mathcal{J}_{\eta|\lambda}(x; q, t) = t^{n((\eta|\lambda)_+)} \sum_R \left(\prod_{\boxed{u}} q^{a+1} t^l \right) \mathcal{G}_{R, (1, 2, \dots, r)}(x; t^{-1})$$

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$$\text{nsH}_{\eta|\lambda}(x; q, t) = \Pi_r \mathcal{J}_{\eta|\lambda}(x; q, t).$$

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Theorem (Blasiak-Haiman-Morse-Pun-S., 2025+)

$$\text{nsH}_{\eta|\lambda}(x; q, t) = \Pi_r \mathcal{J}_{\eta|\lambda}(x; q, t).$$

Follows from

- HHL style formula for both sides.
- Stabilized ribbon LLT's satisfy $\mathcal{G}_{R,(1,2,\dots,r)}(x; t^{-1}) = \Pi_r \mathcal{G}_{R,(1,2,\dots,r)}^{\pm}(x; t^{-1})$.

Raising the flag

- Flagged LLT polynomials serve as a common combinatorial generalization of symmetric LLT polynomials and flagged Schur functions.

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Thank you!